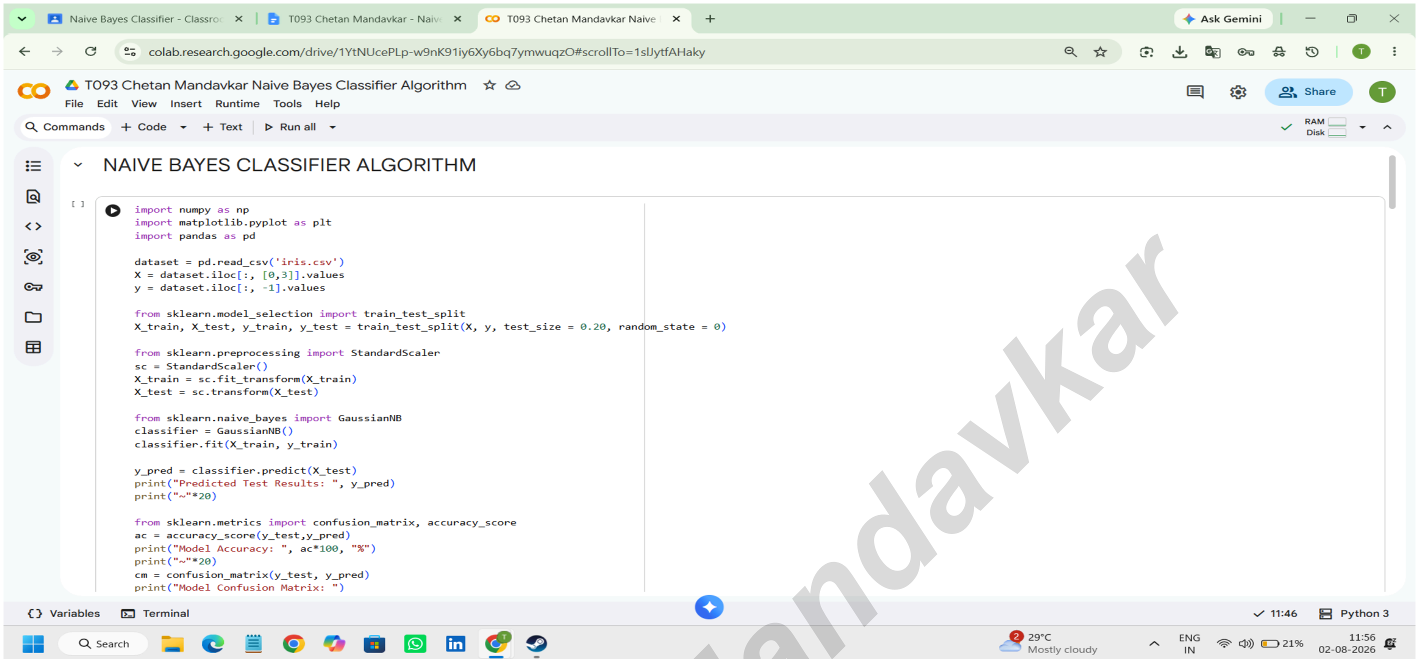
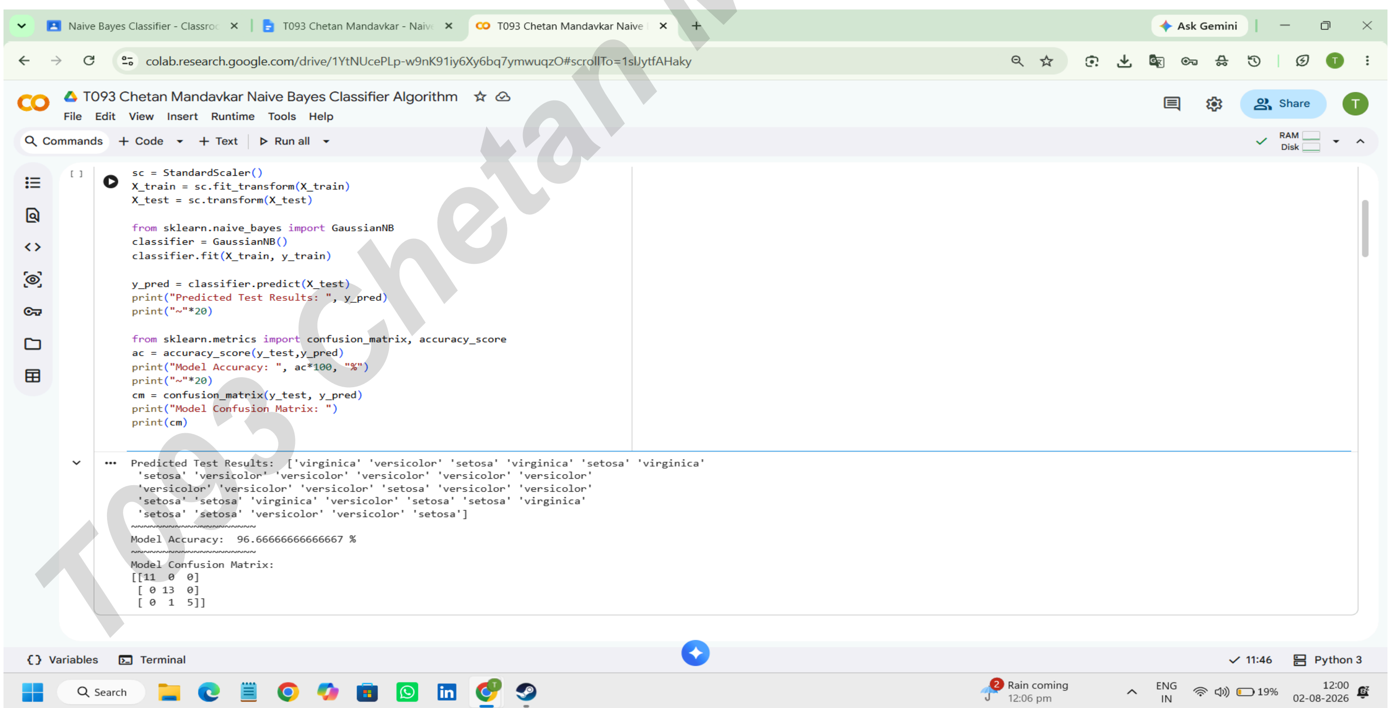
Sheth L.U.J. And Sir M.V. College

Aim:- Naive Bayes Classifier



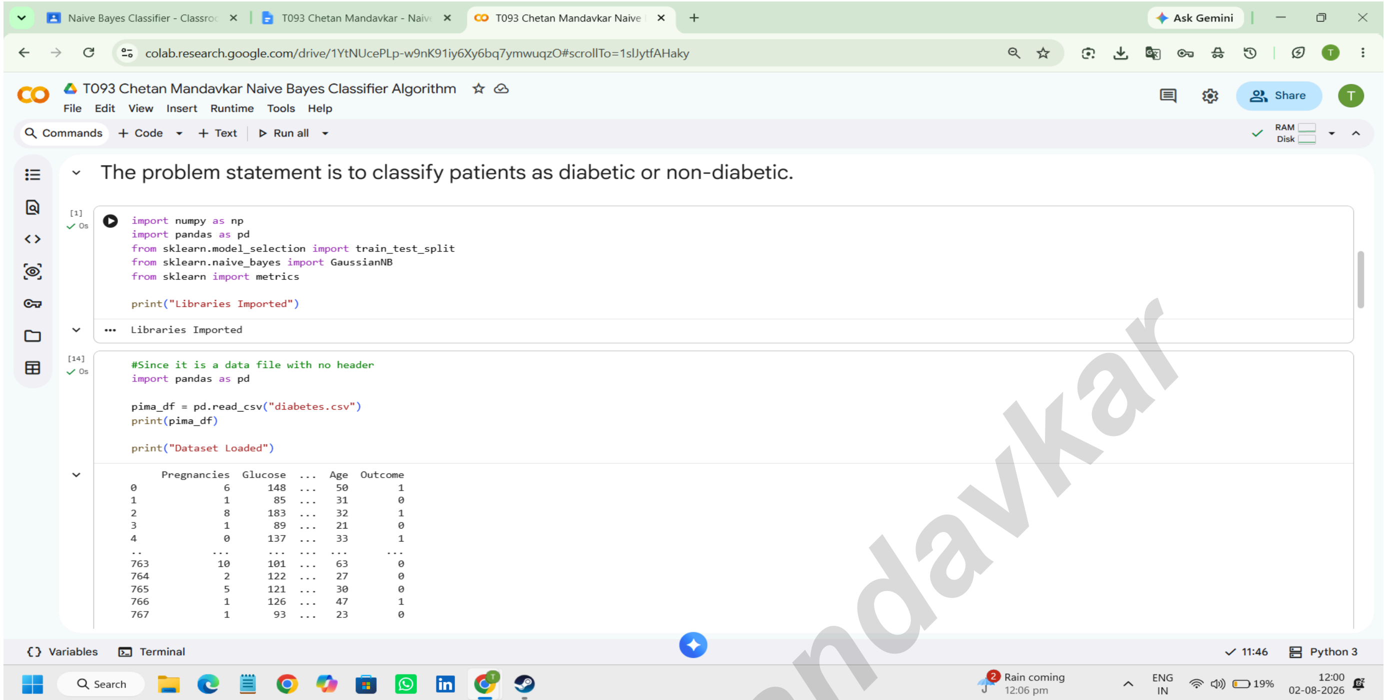


Name:- Chetan Mandavkar

Roll No. T093

Subject:- Artificial Intelligence

Sheth L.U.J. And Sir M.V. College

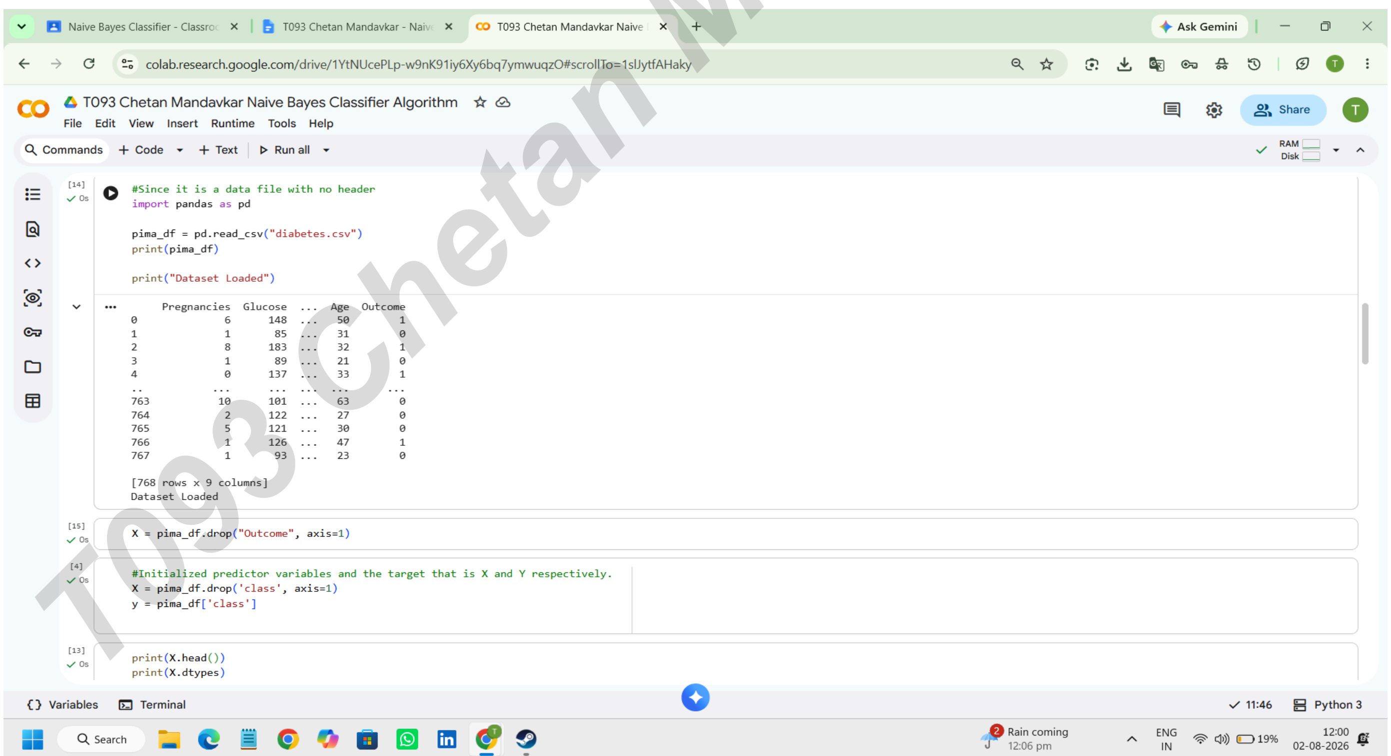



```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn import metrics

print("Libraries Imported")
```

```
Libraries Imported
```

```
#Since it is a data file with no header
import pandas as pd

pima_df = pd.read_csv("diabetes.csv")
print(pima_df)

print("Dataset Loaded")
```

```
     Pregnancies  Glucose  ...  Age  Outcome
0              6      148  ...   50        1
1              1       85  ...   31        0
2              8      183  ...   32        1
3              1       89  ...   21        0
4              0      137  ...   33        1
..           ...      ...  ...  ...      ...
763           10      101  ...   63        0
764            2      122  ...   27        0
765            5      121  ...   30        0
766            1      126  ...   47        1
767            1       93  ...   23        0
```




```
#Since it is a data file with no header
import pandas as pd

pima_df = pd.read_csv("diabetes.csv")
print(pima_df)

print("Dataset Loaded")
```

```
     Pregnancies  Glucose  ...  Age  Outcome
0              6      148  ...   50        1
1              1       85  ...   31        0
2              8      183  ...   32        1
3              1       89  ...   21        0
4              0      137  ...   33        1
..           ...      ...  ...  ...      ...
763           10      101  ...   63        0
764            2      122  ...   27        0
765            5      121  ...   30        0
766            1      126  ...   47        1
767            1       93  ...   23        0

[768 rows x 9 columns]
Dataset Loaded
```

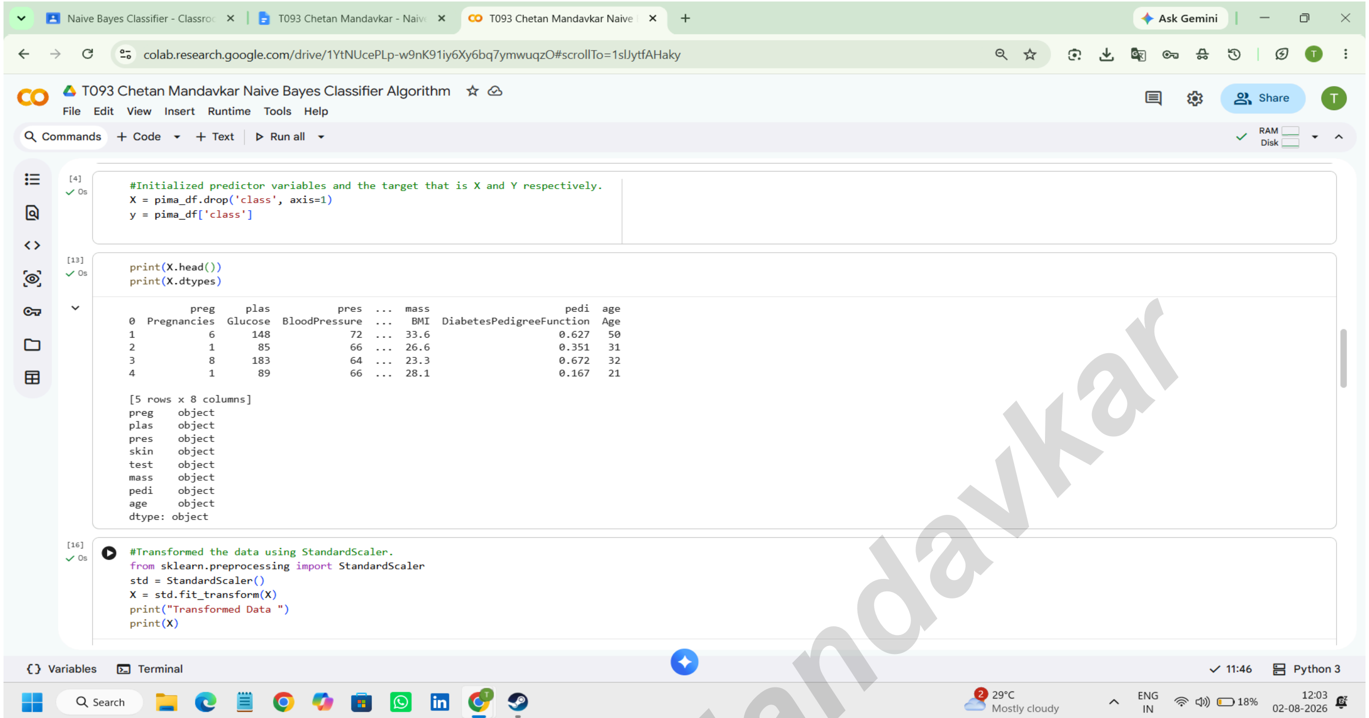
```
X = pima_df.drop("Outcome", axis=1)
```

```
#Initialized predictor variables and the target that is X and Y respectively.
X = pima_df.drop('class', axis=1)
y = pima_df['class']
```

```
print(X.head())
print(X.dtypes)
```


Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence

Sheth L.U.J. And Sir M.V. College



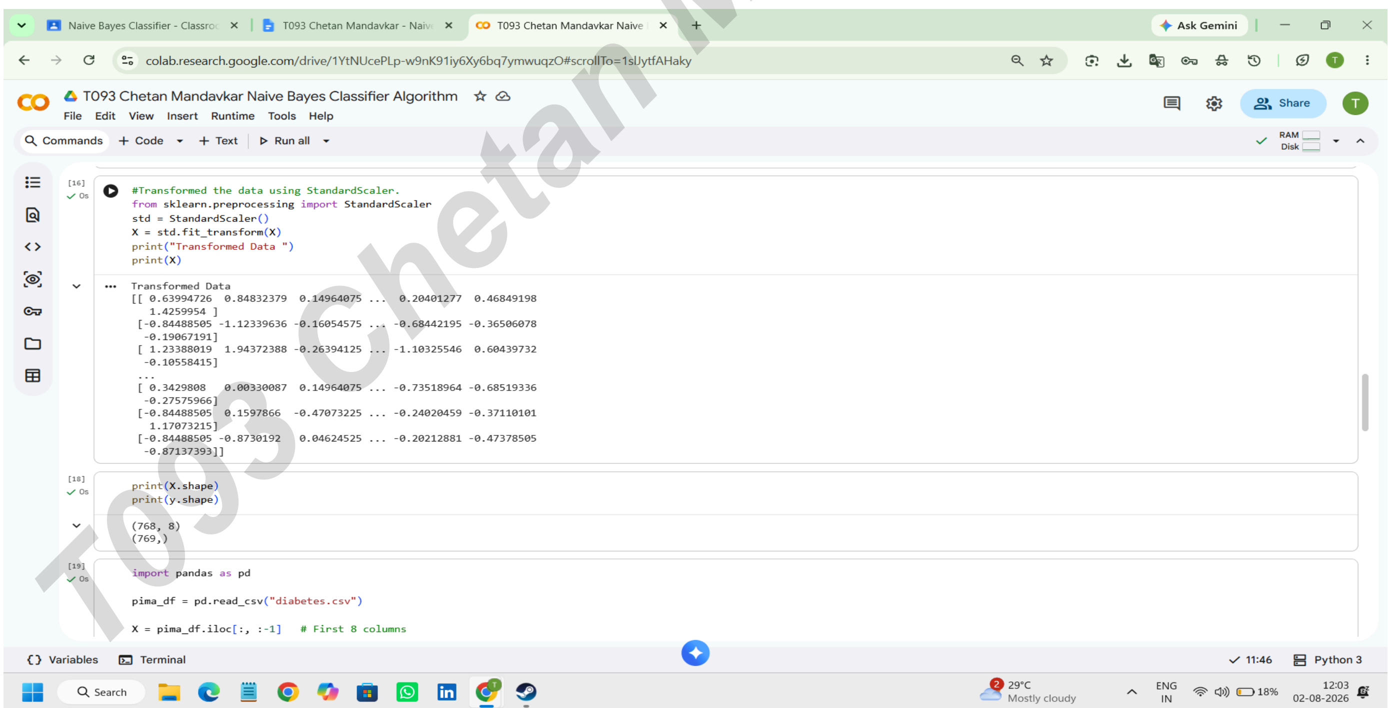
T093 Chetan Mandavkar Naive Bayes Classifier Algorithm

```
#Initialized predictor variables and the target that is X and Y respectively.
X = pima_df.drop('class', axis=1)
y = pima_df['class']
```

```
print(X.head())
print(X.dtypes)
```

```
          preg     plas          pres  ...  mass                      pedi  age
0  Pregnancies  Glucose  BloodPressure  ...   BMI  DiabetesPedigreeFunction  Age
1            6      148             72  ...  33.6                     0.627   50
2            1       85             66  ...  26.6                     0.351   31
3            8      183             64  ...  23.3                     0.672   32
4            1       89             66  ...  28.1                     0.167   21

[5 rows x 8 columns]
preg    object
plas    object
pres    object
skin    object
test    object
mass    object
pedi    object
age     object
dtype: object
```

```
#Transformed the data using StandardScaler.
from sklearn.preprocessing import StandardScaler
std = StandardScaler()
X = std.fit_transform(X)
print("Transformed Data ")
print(X)
```



T093 Chetan Mandavkar Naive Bayes Classifier Algorithm

```
#Transformed the data using StandardScaler.
from sklearn.preprocessing import StandardScaler
std = StandardScaler()
X = std.fit_transform(X)
print("Transformed Data ")
print(X)
```

```
Transformed Data
[[ 0.63994726  0.84832379  0.14964075 ...  0.20401277  0.46849198
   1.4259954 ]
 [-0.84488505 -1.12339636 -0.16054575 ... -0.68442195 -0.36506078
  -0.19067191]
 [ 1.23388019  1.94372388 -0.26394125 ... -1.10325546  0.60439732
  -0.10558415]
 ...
 [ 0.3429808   0.00330087  0.14964075 ... -0.73518964 -0.68519336
  -0.27575966]
 [-0.84488505  0.1597866  -0.47073225 ... -0.24020459 -0.37110101
   1.17073215]
 [-0.84488505 -0.8730192   0.04624525 ... -0.20212881 -0.47378505
  -0.87137393]]
```

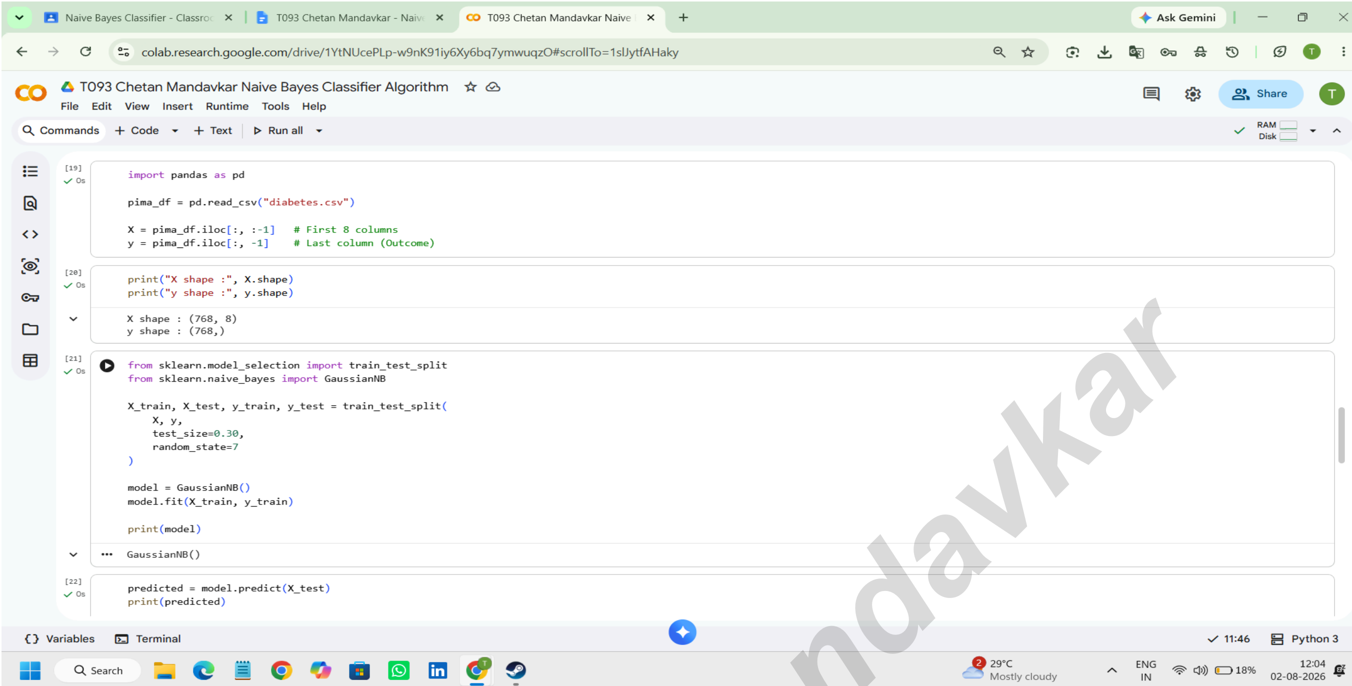
```
print(X.shape)
print(y.shape)
```

```
(768, 8)
(769,)
```

```
import pandas as pd

pima_df = pd.read_csv("diabetes.csv")

X = pima_df.iloc[:, :-1]   # First 8 columns
```

Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence

Sheth L.U.J. And Sir M.V. College



```python
import pandas as pd

pima_df = pd.read_csv("diabetes.csv")

X = pima_df.iloc[:, :-1]   # First 8 columns
y = pima_df.iloc[:, -1]    # Last column (Outcome)
```

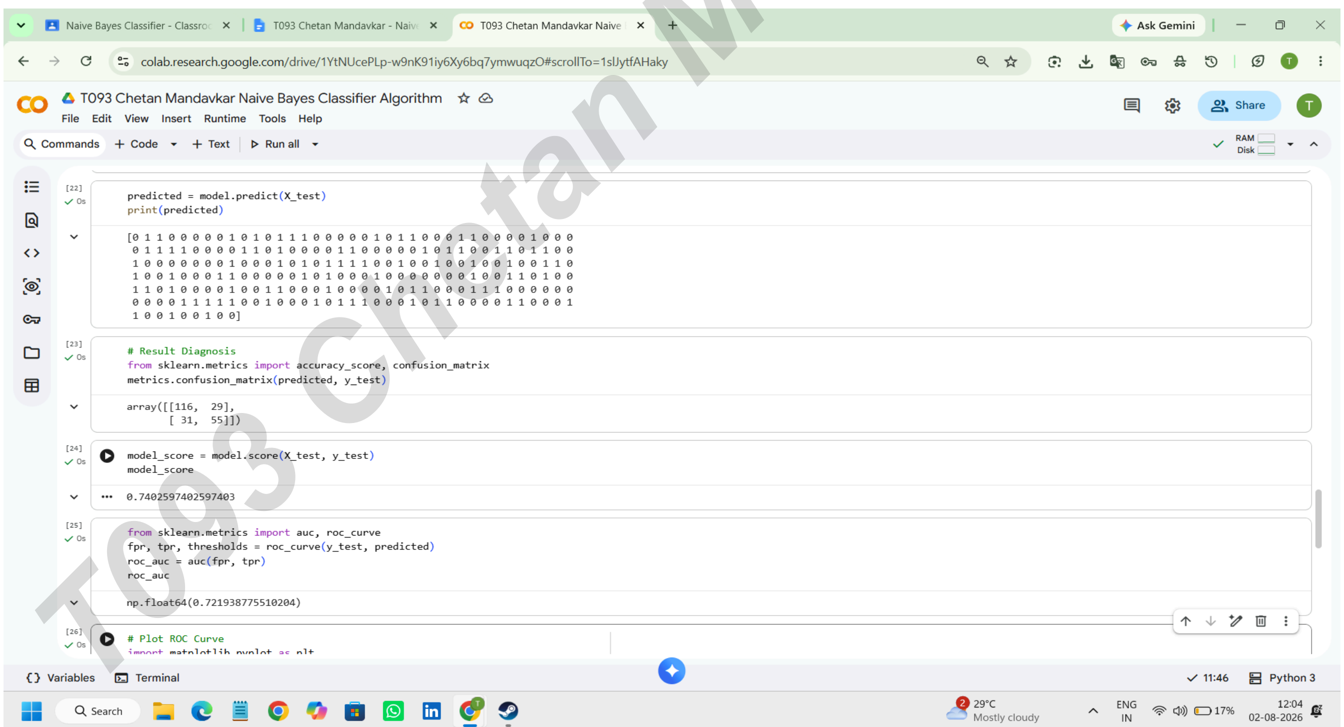
```python
print("X shape :", X.shape)
print("y shape :", y.shape)
```

```
X shape : (768, 8)
y shape : (768,)
```

```python
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.30,
    random_state=7
)

model = GaussianNB()
model.fit(X_train, y_train)

print(model)
```

```
GaussianNB()
```

```python
predicted = model.predict(X_test)
print(predicted)
```



```python
predicted = model.predict(X_test)
print(predicted)
```

```
[0 1 1 0 0 0 0 0 1 0 1 0 1 1 1 0 0 0 0 0 1 0 1 1 0 0 0 1 1 0 0 0 0 1 0 0 0
 0 1 1 1 1 0 0 0 0 1 1 0 1 0 0 0 0 1 1 0 0 0 0 0 1 0 1 1 0 0 1 1 0 1 1 0 0
 1 0 0 0 0 0 0 0 1 0 0 0 1 0 1 0 1 1 1 1 0 0 1 0 0 1 0 0 1 0 0 1 0 0 1 1 0
 1 0 0 1 0 0 0 1 1 0 0 0 0 0 1 0 1 0 0 0 1 0 0 0 0 0 0 0 1 0 0 1 1 0 1 0 0
 1 1 0 1 0 0 0 0 1 0 0 1 1 0 0 0 1 0 0 0 0 1 0 1 1 0 0 0 1 1 1 0 0 0 0 0 0
 0 0 0 0 1 1 1 1 1 0 0 1 0 0 0 1 0 1 1 1 0 0 0 1 0 1 1 0 0 0 0 1 1 0 0 0 1
 1 0 0 1 0 0 1 0 0]
```

```python
# Result Diagnosis
from sklearn.metrics import accuracy_score, confusion_matrix
metrics.confusion_matrix(predicted, y_test)
```

```
array([[116,  29],
       [ 31,  55]])
```

```python
model_score = model.score(X_test, y_test)
model_score
```

```
0.7402597402597403
```

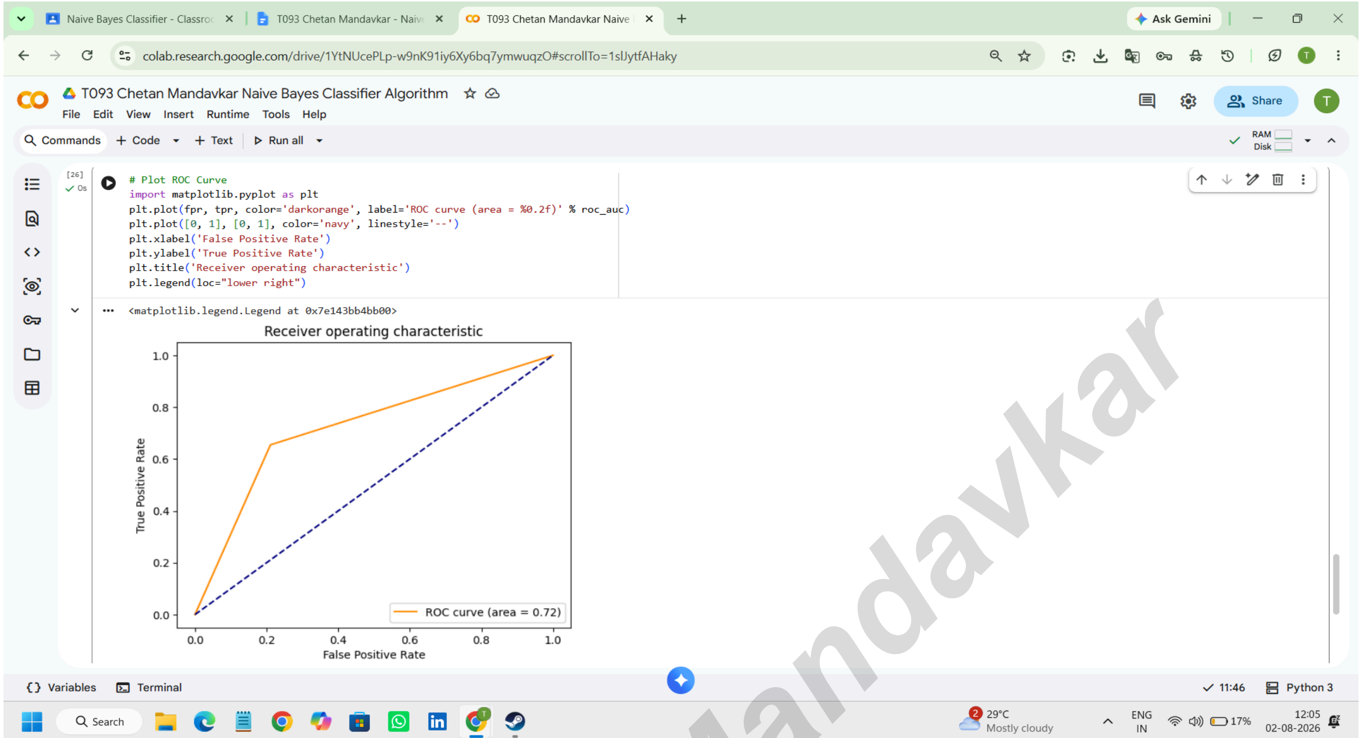
```python
from sklearn.metrics import auc, roc_curve
fpr, tpr, thresholds = roc_curve(y_test, predicted)
roc_auc = auc(fpr, tpr)
roc_auc
```

```
np.float64(0.721938775510204)
```

```python
# Plot ROC Curve
import matplotlib.pyplot as plt
```


Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence

Sheth L.U.J. And Sir M.V. College



Name:- Chetan Mandavkar
Roll No. T093
Subject:- Artificial Intelligence